Gandaki University

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A

Practical Report On

Fundamentals of

Probability And

Statistics

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1. Use multiple regression to examine how Study hours, Attendance, Internet Use, and Sleep Hours influence Final GPA.

a) Which variables significantly predict GPA?

b) What is the R-squared value?

c) Write the regression equation.

d) Interpret the regression coefficient.

e) Calculate correlation coefficient and interpret the result.

Final_GPA	Study_Hours	Attendance	Internet_Use	Sleep_Hours
3.25	24	77	6	8
2.93	19	83	5	7
3.32	11	85	3	5
3.76	16	84	4	8
2.88	12	88	6	5
2.88	19	74	3	8
3.79	7	60	3	8
3.38	18	84	1	6
2.77	21	66	3	5
3.27	8	68	5	7
2.77	22	83	7	7
2.77	12	60	6	5
3.12	8	67	3	7
2.04	6	83	1	7
2.14	10	70	5	5
2.72	14	76	2	8
2.49	8	67	7	5
3.16	22	94	7	8
2.55	16	94	6	7
2.29	6	92	7	7
3.73	14	64	3	7
2.89	8	98	1	6
3.03	18	87	7	8
2.29	20	66	7	6
2.73	19	68	2	6
3.06	12	67	2	5
2.42	18	71	4	6
3.19	12	93	5	5
2.7	20	92	3	5
2.85	17	82	7	6

Ans:

Working Expression:

Statistically Significant if p-value < 0.05

Multiple Linear Regression is used to predict the value of a dependent variable (Final GPA) based on several independent variables (Study Hours, Attendance, Internet Use, Sleep Hours).

Correlation coefficient (r)

$$r = \frac{[n(\Sigma xy) - (\Sigma x)(\Sigma y)]}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2] * [n\Sigma y^2 - (\Sigma y)^2]}}$$

R-Squared (R^2) = 1 - $\frac{SSres}{SStot}$

Regression equation

$$Y = a + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

Where:

Y = Final GPA (dependent variable)

X_1 = Study Hours, X_2 = Attendance, X_3 = Internet Use, X_4 = Sleep Hours

a = constant term

b_i = regression coefficients

Outputs:

Model Coefficients - Final_GPA					
Predictor	Estimate(B)	SE	t	p	Stand. Estimate (Beta)
Constant	2.46985	0.71399	3.459	.002	
Study_Hours	0.00596	0.01599	0.373	.712	0.0715
Attendance	-0.00310	0.00737	-0.420	.678	-0.0784
Internet_Use	-0.04697	0.04103	-1.145	.263	-0.2169
Sleep_Hours	0.12385	0.07233	1.712	.099	0.3210

a.

A multiple linear regression was conducted to predict Final GPA based on Study Hours, Attendance, Internet Use, and Sleep Hours. The results show that none of the independent variables significantly predict GPA at the 0.05 significance level. The p-values for all predictors are greater than 0.05.

b.

Model Summary			
Model	R	R^2	Adjusted R^2
1	0.391	0.153	0.0169
Note. Models estimated using sample size of N=30			

R^2=0.153, meaning 15.3% of the variance of the final GPA is due to the independent variables.

c. Regression Equation:

$Y = a + \beta_1X_1 + \beta_2X_2 + \beta_3X_3 + \beta_4X_4$

$GPA = 2.471 + 0.006X_1 - 0.003X_2 - 0.047X_3 + 0.125X_4$

d. Regression Coefficients:

Interpretation of Coefficients

Study Hours (0.006): A 1 hour increase in study time leads to a 0.006 increase in GPA keeping the other variable constant.

Attendance (-0.003): A 1% attendance increase leads to 0.003 decrease in GPA keeping the other variable constant.

Internet Use (-0.047): A 1 hour increase in internet use tends to lower GPA by 0.047 keeping the other variable constant.

Sleep Hours (0.125): A 1 hour increase in sleep hours improves GPA by 0.125, keeping the other variable constant.

e. Correlation:

Correlation Matrix					
	Final_GPA	Study_Hours	Attendance	Internet_Use	Sleep_Hours
Final_GPA	1				

Study_Hours	0.072	1			
Attendance	-0.067	0.108	1		
Internet_Use	-0.191	0.215	0.118	1	
Sleep_Hours	0.313	0.174	0.090	0.060	1

GPA correlates positively with Sleep hours and Study hours while negatively with Internet use and Attendance.

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2. Use multiple regression to examine how Exercise hours, Sleep hours, Calorie intake, and Stress level influence BMI.

a) Calculate correlation coefficient.
b) What is the R-squared value?
c) Write the regression equation.
d) Estimate the BMI of a person who: Exercises 4 hours/week, Sleeps 7 hours/night, Consumes 2500 calories/day, Has a stress level of 6.
e) Based on the p-values in the coefficients table, identify which independent variables significantly predict BMI. Interpret each significant variable in context.
f) Interpret the coefficients for Exercise Hours and Calorie Intake.

BMI	Exercise_Hours	Sleep_Hours	Calorie_Intake	Stress_Level
18.1	1	5	2252	5
23.1	4	8	2860	4
27.9	9	8	2983	8
23.2	5	7	2814	8
20.9	6	6	1808	7
22.6	3	8	2922	3
26.9	6	5	2615	1
23	7	7	2007	1
27.4	0	8	2954	3
18.8	5	5	2203	6
20.5	7	5	1951	7
22	4	6	1853	6
18.8	3	7	2943	6
35.8	1	7	2427	6
19.6	5	6	2386	3
27.7	5	6	1903	6
21.3	0	7	2053	8
21.9	8	7	2935	2
25.2	5	8	2309	5
18.2	2	8	1898	1
21.3	3	6	1952	1
23.4	3	8	2713	5
20.6	2	5	2677	3
20.7	9	8	2137	4
28.3	2	8	2621	3
28	2	5	2986	1
22.2	3	6	2756	1
23.7	6	5	1960	5

28.9	3	8	2600	6
18.3	8	6	2197	3

Ans:

Working Expression:

Multiple Linear Regression is used to predict the value of a dependent variable based on several independent variables.

Correlation coefficient (r)

$$r = \frac{[n(\Sigma xy) - (\Sigma x)(\Sigma y)]}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2] * [n\Sigma y^2 - (\Sigma y)^2]}}$$

R-Squared (R^2) = $1 - \frac{SSres}{SStot}$

Regression equation

$Y = a + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$

Where:

Y = BMI (dependent variable)

X_1 = Exercise Hours, X_2 = Sleep Hours, X_3 = Calorie Intake, X_4 = Stress Level

a = constant term

$\beta_1, \beta_2, \beta_3, \beta_4$ = regression coefficients

Statistically Significant if $p\text{-value} < 0.05$

Output:

a.

Correlation Matrix					
	BMI	Exercise_Hours	Sleep_Hours	Calorie_Intake	Stress_Level

BMI	1				
Exercise_Hours	-0.183	1			
Sleep_Hours	0.260	-0.062	1		
Calorie_Intake	0.306	-0.162	0.319	1	
Stress_Level	0.082	0.077	0.074	-0.141	1

Interpretation of Correlation

- Exercise_Hours (-0.183): There is a weak negative correlation between Exercise Hours and BMI. As exercise increases, BMI tends to decrease slightly.
- Sleep_Hours (+0.260): There is a moderate positive correlation between Sleep Hours and BMI. Individuals who sleep more tend to have a slightly higher BMI.
- Calorie_Intake (+0.306): This shows the strongest positive correlation with BMI. Higher calorie intake is associated with higher BMI.
- Stress_Level (+0.082): This is a very weak positive correlation.

b. R-squared value:

Model Fit Measures		
Model	R	R^2
1	0.392	0.154
Note. Models estimated using sample size of N=30		

This means that 15.4% of the variation in BMI is due to the independent variables.

c.

Model Coefficients - BMI					
Predictor	Estimate(B)	SE	t	p	Stand. Estimate (Beta)
Intercept	13.67725	5.94748	2.300	.030	
Exercise_Hours	-0.22748	0.29962	-0.759	.455	-0.142
Sleep_Hours	0.56169	0.67168	0.836	.411	0.164
Calorie_Intake	0.00246	0.00199	1.240	.226	0.247
Stress_Level	0.20400	0.33009	0.618	.542	0.116

Regression Equation:

$$Y = a + \beta_1X_1 + \beta_2X_2 + \beta_3X_3 + \beta_4X_4$$

$$BMI = 13.677 - 0.227X_1 + 0.562X_2 + 0.002X_3 + 0.204X_4$$

d. Estimate BMI for:

- Exercise Hours = 4
- Sleep Hours = 7
- Calorie Intake = 2500
- Stress Level = 6

BMI = 13.677 - 0.227(4) + 0.562(7) + 0.002(2500) + 0.204(6)

BMI = 13.677 - 0.908 + 3.934 + 5.000 + 1.224

BMI = 22.93

A person with these lifestyle factors is predicted to have a BMI of approximately 22.93, which falls within the healthy range (18.5–24.9).

e. Based on the p-values in the coefficients table, none of the independent variables significantly predict BMI at the 0.05 significance level.

f. Interpret the coefficients for Exercise Hours and Calorie Intake:

Exercise Hours (-0.227): For every additional hour of exercise per week, BMI is expected to decrease by 0.227 units, holding other variables constant. This supports the idea that exercise helps reduce body fat and lowers BMI.

Calorie Intake (0.002): For every additional calorie consumed per day, BMI increases by 0.002 units, assuming other factors are constant. Though the coefficient is small, over hundreds of calories, this effect becomes meaningful.

3. You are given a dataset of 50 values representing student test scores out of 100. Use the data to explore data visualization and descriptive statistics.

42, 42, 42, 43, 44, 45, 45, 47, 48, 48, 48, 49, 53, 56, 58, 58, 58, 61, 61, 63, 64, 64, 64, 69, 71, 71, 71, 72, 74, 77, 77, 78, 80, 80, 81, 81, 82, 84, 85, 85, 86, 86, 89, 91, 92, 92, 94, 95, 96, 97

a) Construct a stem-and-leaf plot for the given dataset.
b) Calculate the mean, median, mode, standard deviation, and range.
c) Draw a box-and-whisker plot using the dataset.
d) Is the data symmetric, skewed left, or skewed right?
e) What conclusions can you draw about class performance?

Ans:

Working Expression:

Skewness

If Sk(P) = 0, distribution is symmetrical.

If Sk(P) > 0, positive or right skewness.

If Sk(P) < 0, negative or left skewness.

Output:

a. Stem-and-leaf plot:

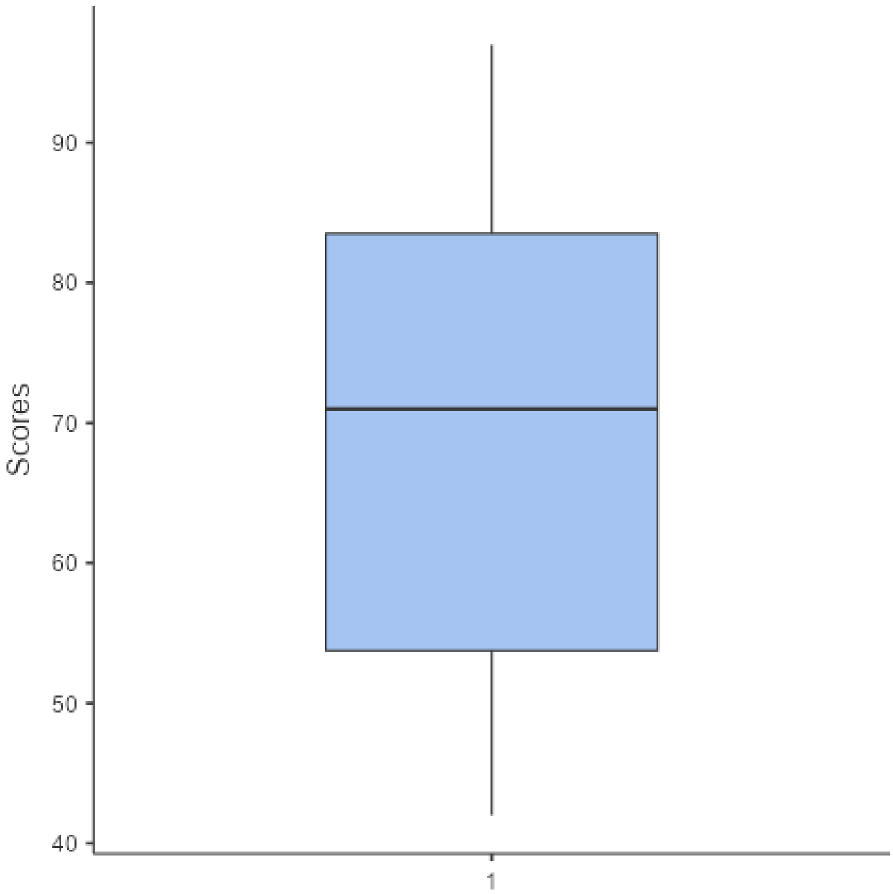
Stem	Leaf	Frequency
4	2 2 2 3 4 5 5 7 8 8 8 9	12
5	3 6 8 8 8	5
6	1 1 3 4 4 4 9	7
7	1 1 1 2 4 7 7 8	8
8	0 0 1 1 2 4 5 5 6 6 9	11
9	1 2 2 4 5 6 7	7

b. Mean, median, mode, standard deviation and range:

Descriptives	
N	50
Mean	68.8
Median	71.0

Mode	42.0[a]
Standard deviation	17.4
Range	55
[a] More than one mode exists, only the first is reported	

c. Box-and-Whisker plot:



d. Skewness:

Descriptives	
	Scores
N	50
Skewness	-0.0760
Std. error skewness	0.337

As skewness is less than 0 i.e. (-0.0760<0), the data is negatively skewed or left skewed.

e. Conclusions

Based on the statistical analysis:

- Overall Performance: The class average of 68.78 suggests moderate performance, depending on the grading scale used.

- Distribution Shape: The nearly symmetric distribution (light left skew) indicates most students performed around the average, with a few lower-performing students pulling the mean slightly below the median.
- Variability: The standard deviation of 17.436 shows considerable spread in student performance, indicating significant differences in achievement levels.
- Range of Scores: With scores ranging from 42 to 97 (range of 55 points), there's substantial variation in class performance.
- Median Performance: Half the class scored above 71, suggesting that while the mean is 68.78, a good portion of students performed reasonably well.